C5 MRTMOS

Illustration of Compensation Networks in CMOS Amplifiers

IEEE Solid-State Circuits Society
Technical Committee on the Open Source Ecosystem
(TC-OSE)

SSCS

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Team Background

Discord name	Name	Affiliation	Experience	Role
madhavgulati277	Madhav Gulati	IIT Delhi	Graduate student	Team lead
ramann_696	Raman Naik	IIT Delhi	Undergraduate student	Team member
tur0922	Turanyaa Garg	IIT Delhi	Undergraduate student	Team member

- We all are IIT Delhi students, having a decent experience in analog and RF circuits. We have taken various advanced courses on Analog Integrated Circuits, compact modeling of Semiconductor devices, CMOS RF design, Analysis and Design of operational amplifiers.
- We also have a decent experience in designing and layout of cells (memory cells, adders, RF circuits, amplifiers) using Cadence tools as well as open source tools (Xschem, Magic and NGSpice)

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Progress tracker

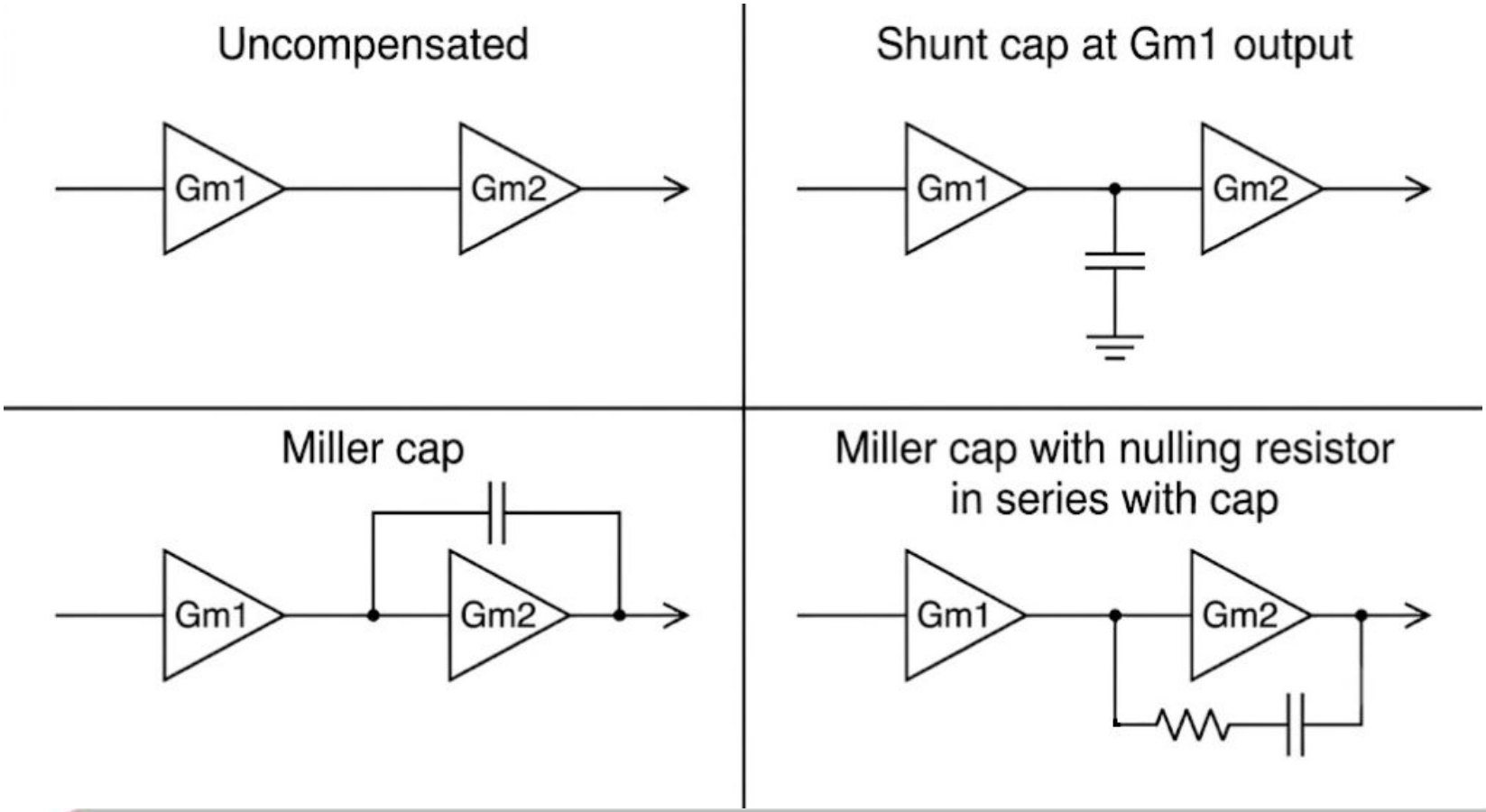
- Remarks by Prof. Peter Kinget (proposal review, comments on Github issue):
  - ~~make a **full pin-out list** including all your control pins~~ - **Done**
  - ~~make a **complete control table** to show how you will be able to cover all your desired experiments~~ - **Done**
  - work out a set of concrete configurations that you want to use the chip in -- what experiments will they enable - **In progress**
  - ~~provide details for the amplifier schematics~~ - **Done**
  - work through the **numbers** -- what component sizes will you need, will they fit on chip ... and run feasibility simulations using ideal gm, r, and c elements. - **In progress**
  - ~~OTA/Opamps by themselves are very hard to measure .... think of some applications where to use the opamps in that would specifically benefit from the different compensation techniques.~~ - **Done**
  - ~~think of experiments that students have to *identify* which compensation technique has been used, e.g. by looking at features in the step response~~ - **Done**
  - **Progress Tracker (Gantt Chart):** https://1drv.ms/x/c/b67b5a4b19d66dd6/IQCFt5GGdvOgSYuXGDtOKIz9AV6i4Vr8B8j_q4YCoH26yWM

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What are we doing, and why?

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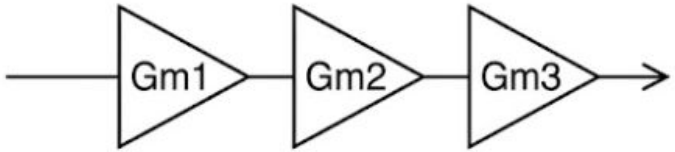
Two stage compensation techniques



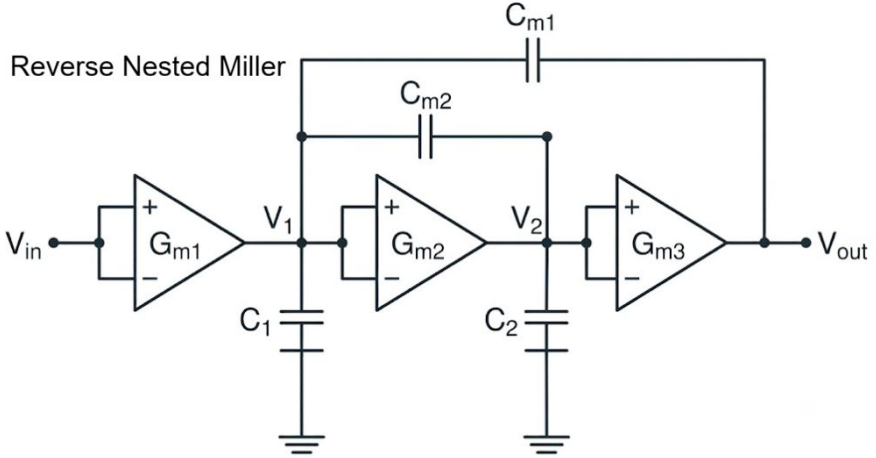
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Three stage compensation techniques

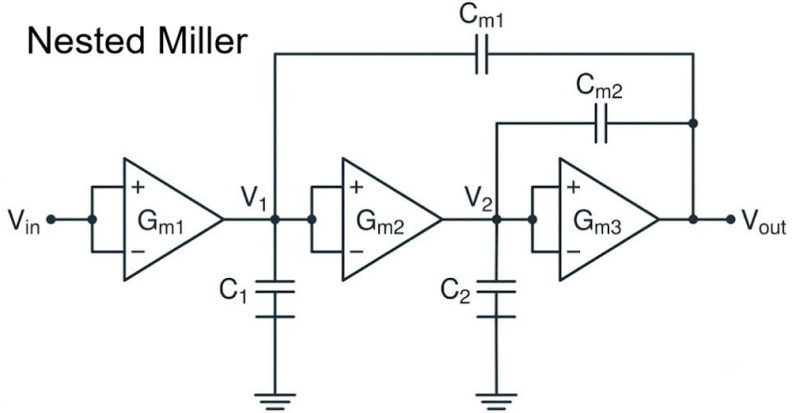
Uncompensated



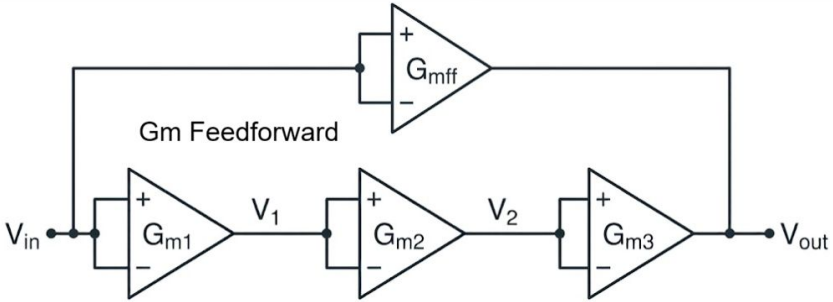
Reverse Nested Miller



Nested Miller



Gm Feedforward



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Schematics for all the stages

- **All Stages and Feedforward:** 5-T OTA
  - We can adjust parameters (width, length, tail current) depending on the use-case!
  - It can be used to invert as well as non-invert!
  - Only 5 transistors needed!
- **Voltage buffer:** Source follower
  - Noninverting, reliable buffer

Assumptions and convention

- For single-input amplifiers, the transconductance $G_m = I_{out}/V_{in}$ would apply.
- For 2-input amplifiers, the transconductance $G_m = I_{out}/(V_+ - V_-)$ would apply.
  - **V_+ and V_- refer to the input signal, not the amplifier input terminals.**
  - G_m would adjust and change sign, but a positive current means current charging the capacitor at the output (coming out of amplifier).
  - This is for algebraic simplicity.
  - This is under the condition that the amplifier has only one output.
- At every node, KVL and KCL will prevail along with the sign convention.

Why we picked them?

- **5-T OTA for all stages and feedforward:**
  - Standardized topology makes layout reuse and verification effortless.
  - Dual inputs allow toggling between inverting and non-inverting.
  - Self-biasing active load and tail current source eliminates complex feedback circuits (common mode feedback circuitry in this case).
  - Decent (roughly 30-40dB) gain via high output resistance with low DC headroom.
  - Uniform architecture simplifies much of our work and student's work!
  - Easy to scale output driving current, gain etc. by adjusting transistor parameters.
    - by scaling W proportionally with I_D's while holding V_{ov} fixed via g_m/I_D.
- **Common drain amplifier as voltage buffer:**

$$Z_{in} \approx \frac{1}{j\omega[C_{gd} + C_{gs}(1 - A_v)]}$$

  - Insulated gate provides very high input impedance, protecting stage gain.
    - at AC, $C_{gs} \approx 30fF$ gives $|Z| \approx 5.3M\Omega$ at 1MHz, still $>> R_{out}$ which is of the order of $50\,k\Omega$
  - Blocks the feedforward current path through the Miller cap connected in series, kills the RHP zero and preserves the feedback path (as output faces the high input impedance and the intermediate output faces the low output impedance of the buffer).

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Some alternatives that came to our mind...

- **Common Source Amplifier for stage 3:**
  - **Temptation:**
    - **Maximum slew rate:** Lacks a tail current bottleneck, pulling massive transient surges directly into the capacitive load.
    - **Ultimate voltage swing:** Only two stacked transistors between supply rails allows near rail-to-rail output.
  - **Why 5-T OTA wins over it?**
    - **Polarity lock-in:** Common Source is permanently inverting, meaning that we won't be able to realize an amplifier in noninverting configuration, when we need it for nested and reverse-nested miller compensation.
- **Folded Cascode OTA (Cascode based circuits in general):**
  - **Temptation:**
    - **Massive single-stage gain:** Stacking transistors multiplies output resistance $(Z_{out} \approx g_m r_o^2)$, for the middle stage (the engine of the 3-stage amplifier). This further boosts the gain.
    - **Superior input range:** Allows input signals to swing close to the rails without pushing devices out of saturation.
  - **Why 5-T OTA wins over it?**
    - **Headroom starvation:** Folded cascode improves headroom over telescopic but still costs more than plain 5T-OTA, plus needs an added folding-bias branch (extra power/complexity).
    - **Biasing fragility:** Requires 3-4 precisely matched bias voltages, where any mismatch carries a potential risk to push cascode devices out of saturation given the already-tight margins

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Some alternatives that came to our mind...

- **Resistive load common source differential pair with tail current**
  - **Temptation:**
    - **Perfect linearity:** Resistors provide frequency-independent, highly linear load impedance.
  - **Why 5-T OTA wins over it?**
    - **Ohm's law trap:** Pushing Stage 3's 1 mA tail current through a 20 $k\Omega$ resistor demands 20 V of headroom. Impossible on our chip. If we reduce the current, it would place a bottleneck on the slew rate. Shrinking the resistor to fit a 3.3 V supply drops output impedance so low that voltage gain degrades sharply as a consequence of output impedance degradation.
- **Current load common source differential pair with resistance in tail.**
  - **Temptation:**
    - **Built-in linearity:** Tail resistor adds local negative feedback (source degeneration), stabilizing transconductance against process shifts.
    - **No bias distribution:** Eliminates routing a precise DC bias voltage (V_{bias}) to the bottom of the stage.

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Some alternatives that came to our mind...

- **Current load common source differential pair with resistance in tail.**

$$A_{dm} = \frac{1}{2} g_m R_D \qquad A_{cm} = \frac{-g_m R_D}{1 + 2 g_m R_{tail}} \qquad A_{cm} \approx -\frac{R_D}{2 R_{tail}}$$

  - **Why 5-T OTA wins over it?**
    - **CMRR destruction:** Good noise rejection requires infinite tail impedance; a physical resistor lets supply noise bleed into the signal.
    - **Throttled slew rate:** A very large tail resistor sized to keep quiescent current low cannot supply more than $(V_{supply} - 2V_{ov})/R_{tail}$ (or even more if we use a resistive load) during large-signal slewing without exceeding available headroom.
      - This caps achievable I_{tail} below the $SR = I_{tail}/C_c$ target, unlike an active current source (~70kΩ output impedance) which holds current roughly constant regardless of voltage, up to compliance.

- **Class AB push-pull output:**
  - **Temptation:**
    - **Small quiescent current:** Burns very less DC current while resting, but opens floodgates during transient load demands.
    - **Thermal efficiency:** Eliminates burning a continuous, chip-heating 1 mA+ DC tail current just to maintain slew rate. Slew rate is decoupled to the tail current here because the path from load to Vdd/ground is the only path conducting current.
  - **Why 5-T OTA wins over it?**
    - **Thermal runaway:** Heating increases temperature, which decreases V_{th}, which increases V_{ov}, which increases quisscient current, which increases power dissipation, which increase temperature further.
      - At high quisscient current and overdrives, when both devices are ON simultaneously, they conduct from supply to ground. **Defeats the purpose of using a class AB push-pull output!**
    - **Single I/O:** We can't connect it in both inverting and noninverting configuration. We need both to realize all our experiments.

Some alternatives that came to our mind...

- **5-T OTA with unity-gain feedback (for the Miller voltage buffer):**
  - **Temptation:**
    - **Topology reuse:** We already have the 5-T OTA block designed, biased, and verified. Copy-pasting it as our buffer saves layout and verification time.
    - **Precision voltage tracking:** The high open-loop gain of the 5-T OTA guarantees the closed-loop buffer tracks the voltage almost perfectly $(A_v \approx 1, A_{cl}=1\text{-}1/A_{ol})$, better than Source Follower.
  - **Why Source Follower wins over it?**
    - **The "Loop-in-a-Loop" danger:** The Miller feedback path must be very fast to stabilize the main amplifier. A 5-T OTA introduces its own internal poles, adding severe phase lag into the compensation path and destroying the main phase margin.
    - **Self-stability nightmare:** A 5-T OTA configured in unity-gain requires its own compensation to prevent oscillation. A source follower is inherently broadband and significantly stable.
    - **Area and power bloat:** Using 5 transistors and a dedicated tail current source just to block feedforward current is an overkill. A 1-2 transistor Source Follower provides the exact unidirectional isolation we need with a fraction of the silicon area and capacitance.

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An improvement: Using a 6T OTA for Gm1

- The first stage of the 3-stage OTA is designed to filter out the input noise from the raw input signal, i.e. to suppress common mode interference.
- The common mode gain is inversely proportional to the tail resistance, and the CMRR is directly proportional to the tail resistance
- Yes, there is a cascode but that is not that cumbersome to bias compared to a cascode amplifier or folded cascode amplifier.
  - We can directly connect the beta multiplier with the low voltage cascode current mirror, and the 2 stacked NMOS in the right arm of the current mirror can be reused as the tail transistors of the 6T OTA.
  - We don’t need any extra bias voltages. It is a plug and play circuit.
  - For biasing a folded/telescopic cascode amplifier the way we are biasing the 6T OTA, we would need to copy the current **twice** using **two** separate current mirrors.
    - We would have to ensure that the two current mirrors have perfectly matching currents, requiring a common-mode feedback circuitry.

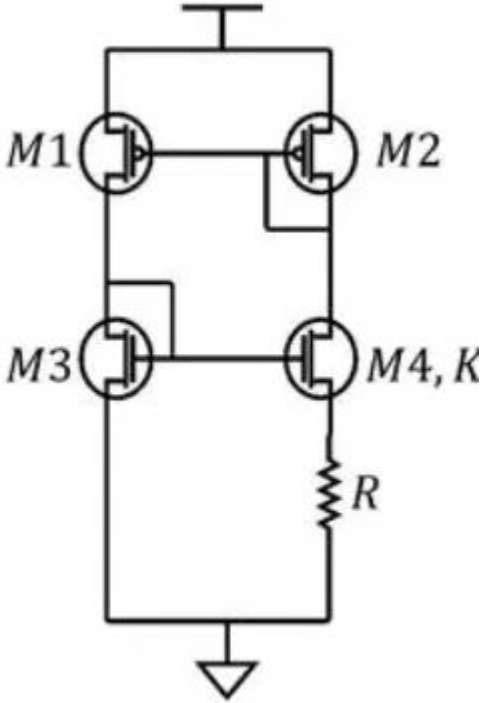
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Biasing of the Amplifiers

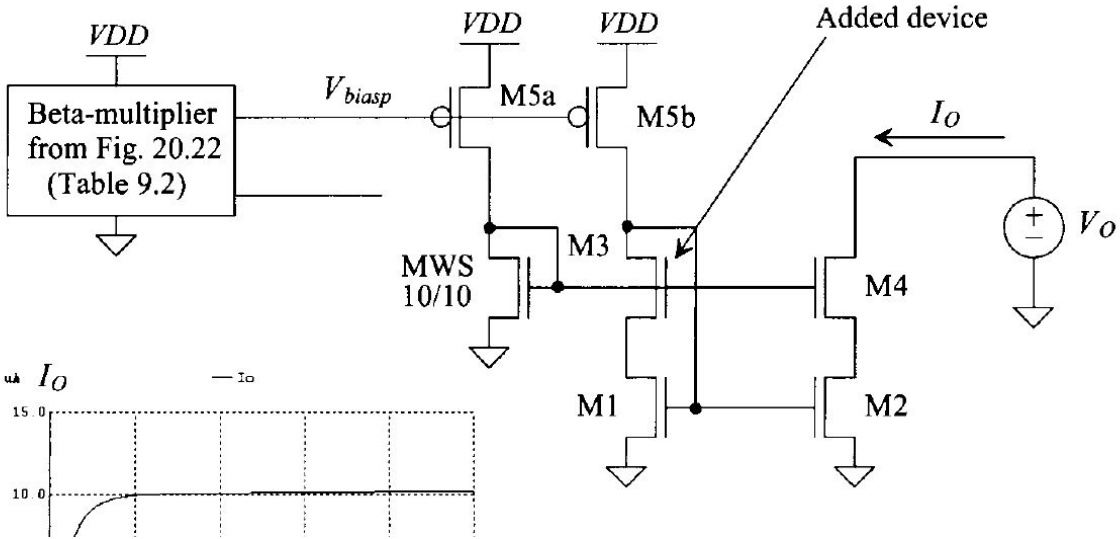
- As we are taping out, we have to be **doubly cautious and careful here.**
- We will be distributing currents from a single beta multiplier circuit and current mirrors (Beta multiplier resistance is connected externally via pin so that it can be tuned)
  - Low voltage cascode (Sooch current mirror) for biasing the 6T OTA
  - Direct current mirror (connecting diode connected NMOS to tail NMOS) for 5T OTA.
  - For the Gmff, we are using a diode connected NMOS and externally connected resistor, as we would want maximum flexibility in it.
- For the input nodes, we won't be putting any bias circuitry. The oscilloscopes have an inbuilt offset voltage feature, it would be better to use that only. Else, we would end up making a very heavy and complex circuitry or using a simple voltage divider and capacitor based circuit, which is very prone to errors and PVT variations which we won't want.

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Circuits used for biasing

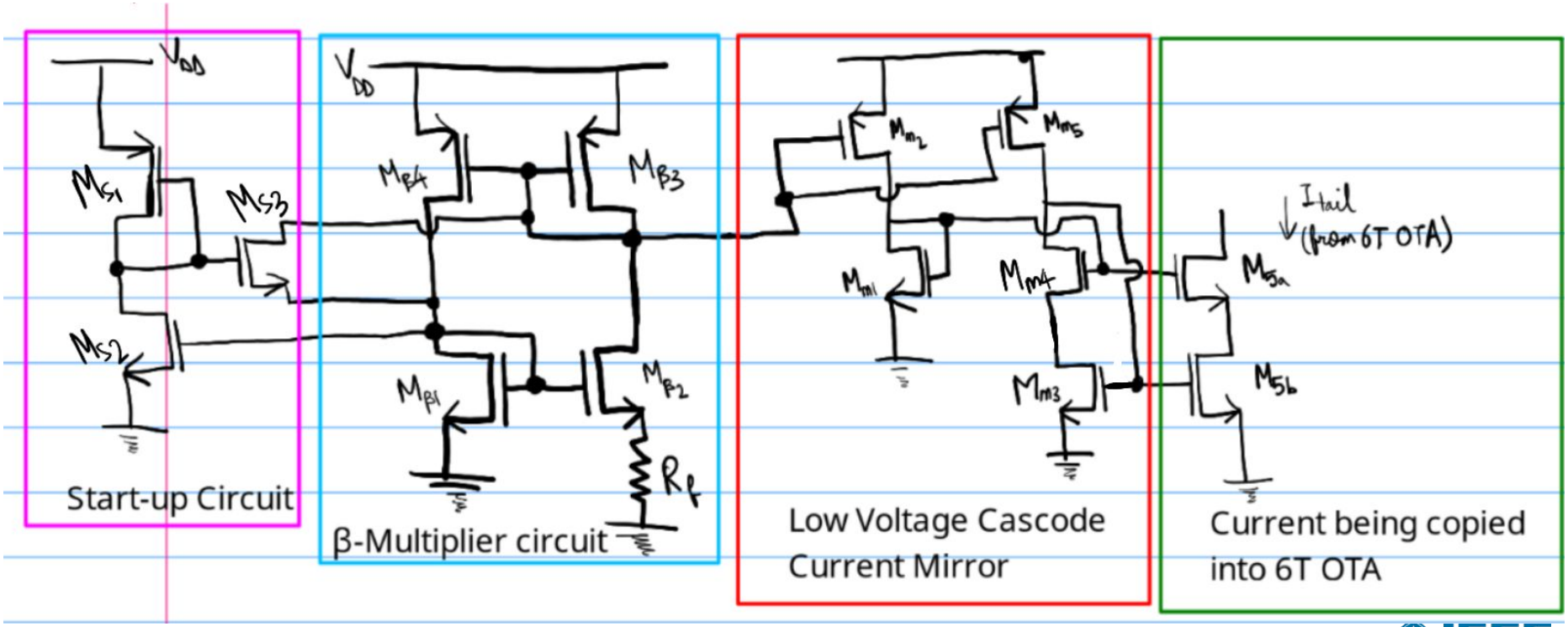


Beta Multiplier Circuit
Source: cmosedu.com (Prof. R. Jacob Baker)



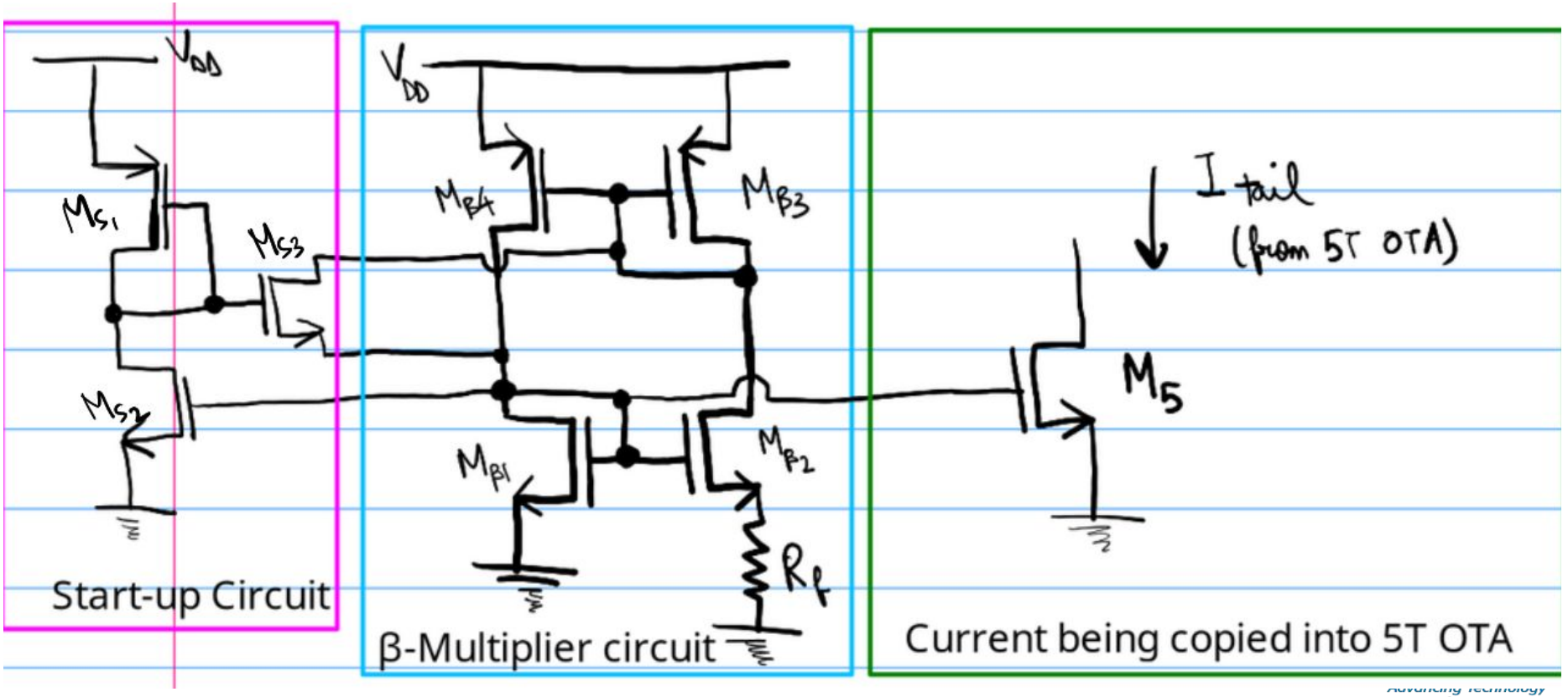
Low Voltage Cascode Current Mirror
Source: CMOS Circuit Design, Layout and Simulation, 3rd ed.

Biasing 6T OTA

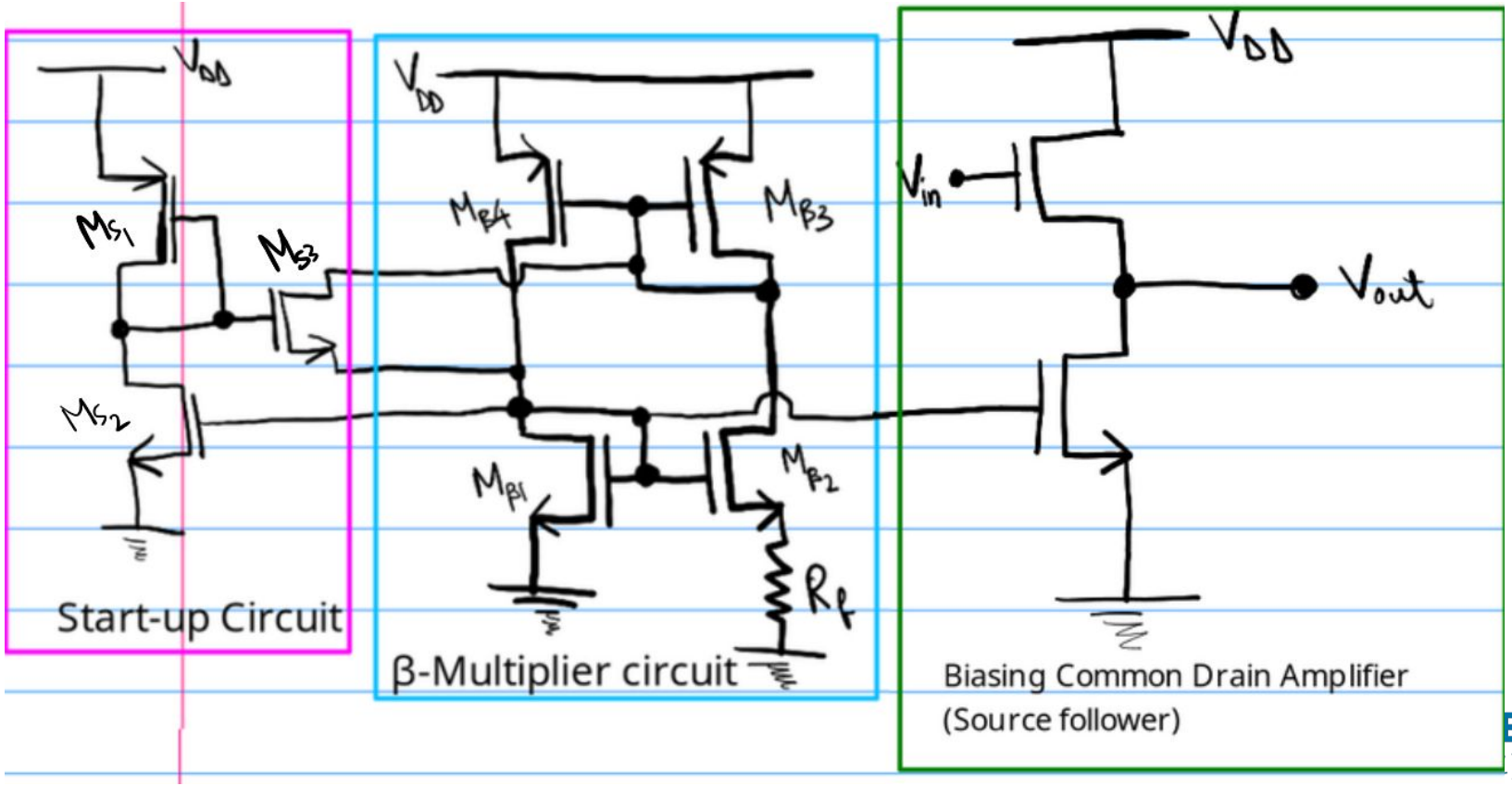


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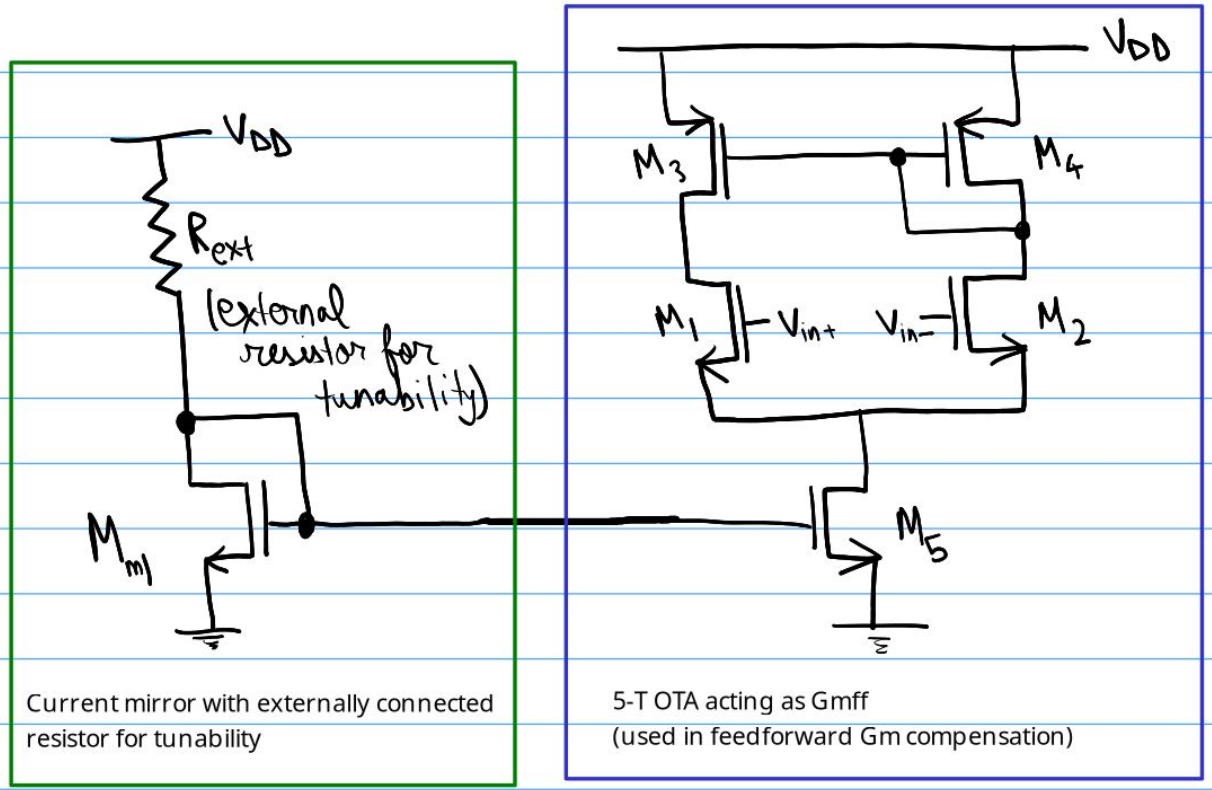
Biasing 5T OTA



Biasing Source follower circuit



Biasing Feedforward Amplifier (Gmff)



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Should R/C be on/off chip?

Resistor

- Nulling resistors must be placed off-chip.
- Students sweep to empirically discover the exact value.
- On-chip resistors lack any tuning capabilities.
  - Even if we realize blocks of resistors (ohms per square), they are prone to inaccuracies and discreteness in values.
- Wiper resistance uncertainty stays below 5% worst-case.
- Fixed precision metal-film resistors serve secondary verification.
- Precision resistors provide an ultra-tight 0.1% tolerance.
- On-chip resistor variations degrade matching by 20%.
- Off-chip components close the hand-analysis loop.
- Another option: Potentiometers enable continuous sweeping of R_{null}.

Capacitor

- All compensation capacitors must be off-chip components.
- On-chip MOM density is roughly $2\text{fF}/\mu\text{m}^2$.
- Target $C_c = G_{m1}/(2\pi \cdot \text{GBW}) \approx 318\ \text{pF}$.
- Area needed is $159{,}000\mu\text{m}^2$ per capacitor.
- Total available chip area is $1{,}250{,}000\mu\text{m}^2$!
  - Caps will take away atleast a tenth of the area.
  - Very hard to tune the caps on-chip, if we use varactors.
- On-chip integration is physically completely impossible given that there are other components as well.
- Off-chip C0G dielectric guarantees $\pm 0.1\%$ tolerance.
- On-chip process variations suffer a $\pm 20\%$ error.
- Precise values allow exact experimental verification.

The final verdict

- Keep it off-chip!
- Let students decide the values and plug them in!

A Problem: Breadboard parasitics

- Breadboards are notoriously hostile to high-frequency analog integrated circuits.
- The physical structure of a standard breadboard introduces roughly 2 pF to 5 pF of hidden parasitic capacitance between adjacent rows.
- Furthermore, standard jumper wires act as tiny antennas, introducing stray inductance and coupling ambient noise directly into sensitive paths. It is a major error for high frequency circuits.
- How does this jeopardize our compensation experiments?
- These invisible environmental parasitics artificially shift the dominant and non-dominant poles before a student even plugs in a compensation capacitor.
- An intentionally uncompensated, unstable OTA might not oscillate simply because the breadboard itself is accidentally acting as a compensation network!
- High-impedance internal nodes routed off-chip will suffer severe phase margin degradation entirely due to the test setup. at compensation node, even 5pF of stray capacitance creates a parasitic pole at 318kHz, within an octave of our 482kHz GBW target.
- The exact, predictable mathematical derivations we want students to empirically verify (like precisely shifting the RHP zero) become completely muddled by environmental noise.
- **The bottom line: Students must measure the exact physics of our silicon, not the physical flaws of a plastic breadboard.**

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Solution: PCB implementation (microcontroller-style board)

- We will bypass the breadboard entirely and design a fully dedicated, professional-grade evaluation PCB.
- This shifts the project from a fragile "student chip" into a robust, self-contained educational microcontroller-style platform.
- To tackle the parasitic capacitance, we must execute a careful ground plane strategy.
- The Trap: Completely removing the ground plane beneath the entire SoC is dangerous; it destroys the return paths for digital SPI control logic, causing digital switching noise to radiate directly into our analog stages.
- The Real Solution: We will maintain a solid ground plane for power delivery and digital logic, but utilize highly localized "copper keep-outs" (voids).
- We will strictly remove all ground and power planes only directly beneath the sensitive high-impedance routing traces and the female plug-in receptacles.
- **We will use our knowledge and experience in PCB design and routing :)**

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Lab experiments

Sequence and Methodology

- **Phase 1: Single-Stage Characterization (Small & Large Signal)**
  - Measure discrete amplifier blocks
  - Small signal to measure Gm, Large signal to measure swing and ICMR
  - Tail current to measure power $(P_{diss} = V_{DD} * I_{tail})$
- **Phase 2: Frequency response, compensation (100mV p-p input)**
  - Compare frequency response in uncompensated and various forms of compensation
  - Vary values of R/C or feedforward Gm in compensation networks, and plot graphs to develop intuition.
- **Phase 3: Characterizing multi-stage amplifiers (small & large signal)**
  - Measure the entire amplifier combined, steps similar to phase 1
- **Phase 4: Basic applications and system level applications**
  - Filter design

Measuring and Characterization of single OTA: Gain and output impedance

- Students would connect the amplifier in negative feedback network configuration. Input voltage should be small (50mV peak-peak or so) at a reasonable frequency (5-10kHz).
- The OTA would be connected to a resistive load, and the voltage induced in the resistor would be fed back into the inverting terminal of the OTA. Input voltage would be connected to the noninverting terminal.
- By varying the resistive load R_L, students would in turn, vary the negative feedback gain.

$$A_v = \frac{V_{out}}{V_{in}} = \frac{G_m \cdot R_L || R_{out}}{1 + G_m \cdot R_L || R_{out}} \implies \frac{1}{A_v} - 1 = \left(\frac{1}{G_m}\right)\frac{1}{R_L} + \frac{1}{G_m R_{out}}$$

- Plotting $(1/A_v)$-1 vs $1/R_L$ yields a straight line where the slope directly equals $1/G_m$. The y-intercept comes out to be $1/G_m \cdot R_{out}$.
- It is recommended to use small values of resistances (upto a few $k\Omega$) to first measure G_m as a small value of R_L would also push the output pole far away.
- At high values of R_L, we can end up measuring R_{out}!
- But at very high values of R_L, the output pole would occur at lower frequencies.
- After experiment is done, students would get the passive and transistor parameters and topologies of amplifiers, and measure via hand calculation!

Measuring Gain and Characterization of the Voltage buffer

- It can be done in a straightforward way, as the voltage gain of the voltage buffer is constant and there is no fear of excess voltage and voltage gain deterioration depending on the load unlike in an OTA.
- Connect a sinusoidal voltage source at the input of the voltage buffer. 50mV peak-peak
- We can measure the output voltage and calculate transfer function.
- We can vary the frequency and amplitude to get an idea of the output swing and bandwidth.
- We can also get to know the ICMR, as we increase the input voltage and we can get to know deviation in output waveform (eg. Clipping) to identify ICMR limits.
- After experiment is done, students would get the passive and transistor parameters and topologies of amplifiers, and measure via hand calculation!

Compensation Networks – Analysis and comparison

1) **Establish the Baseline:** Students utilize their previously extracted G_m and R_{out} values for the individual blocks.
2) **The Uncompensated Benchmark:** Students cascade the blocks into a 2- or 3-stage amplifier and measure the raw frequency response (identifying poles, zeros, DC gain, and phase/gain margins).
3) **Calculate, Then Connect:** Students mathematically design various compensation networks by hand before physically wiring them on the bench. Make the measurements as in step 2.
4) **Observe and Compare:** Students populate a master observation table to directly compare how each compensation topology shifts the parameters and stabilizes the system.

Analysis and Characterization of multi-stage amplifiers

- Students would connect the multi-stage OTA in negative feedback network configuration. Input voltage should be small (50mV peak-peak or so) at a reasonable frequency (5-10kHz). **Don't connect any compensation network yet for baseline measurements.**
- The OTA would be connected to a resistive load, and the voltage induced in the resistor would be fed back into the inverting terminal of the OTA. Input voltage would be connected to the noninverting terminal.
- By varying the resistive load R_L, students would in turn, vary the negative feedback gain.

$$A_v = \frac{V_{out}}{V_{in}} = \frac{G_m \cdot R_L || R_{out}}{1 + G_m \cdot R_L || R_{out}} \implies \frac{1}{A_v} - 1 = \left(\frac{1}{G_m}\right)\frac{1}{R_L} + \frac{1}{G_m R_{out}}$$

- Plotting $(1/A_v)$-1 vs $1/R_L$ yields a straight line where the slope directly equals $1/G_m$. The y-intercept comes out to be $1/G_m \cdot R_{out}$.
- It is recommended to use small values of resistances (upto a few $k\Omega$) to first measure G_m as a small value of R_L would also push the output pole far away.
- Compare the experimentally extracted overall parameters (G_m, R_{out}) of the complete multi-stage amplifier against the individual stage parameters previously measured in phase one. Verify the theoretical mathematical relationships of the cascaded stages.

Vary compensation network elements, develop intuition

- **Calculate by Hand:** Derive the transfer function A_v to map out target pole and zero locations.
- **Connect the Topology:** Link the multi-stage amplifier network to your chosen compensation loop configuration.
- **Inject the Step Response:** Vary R,C and feedforward G_m values, and apply a step response to observe the resulting waveforms, and observe the I/O plots and make an observation table.
- **The Transient-to-Control Theory Map:**
  - **Sluggish Rise (No Overshoot):** Overdamped response ($\zeta>1$) with excessive Phase Margin ($\varphi_m>76^\circ$) but choked bandwidth.
  - **Ringing & Peak Overshoot:** Underdamped condition ($0<\zeta<1$); non-dominant poles have crept inside the loop bandwidth, degrading Phase Margin ($\varphi_m<60^\circ$ typically).
  - **Sustained Oscillations:** Total system instability where Phase Margin hits 0° and loop poles have migrated into the unstable Right-Half Plane (RHP).
  - **The "Slow Creeping Tail":** Diagnostic signature of a Pole-Zero Doublet; imperfect pole-zero cancellation traps an agonizingly slow exponential time constant in your settling path.
  - **Initial Reverse-Direction "Dip":** Classic Non-Minimum Phase behavior caused by a dominant Right-Half Plane (RHP) Zero momentarily outrunning the main amplifier path.
- **First do the calculations and predict the waveforms, then do it practically. Don’t be a simulator monkey!**

System level applications 1: Design a filter!

- You have the specifications of all the amplifiers (discrete and combined!)
- Most amplifiers have isolated terminals.
- Design a filter with 1-2 amplifiers:
  - State variable, Multiple feedback, Sallen Key, Biquad, simple amplifier based active filters.
  - Bandwidth is an important aspect in filter design, and poles/zeroes and therefore, compensation play a significant role.
- First calculate the transfer function by hand.
- Then form the circuit and connect.
- After that, you can verify the measurements with experimental results.

System level applications 2: Active inductor!

- You have the specifications of all the amplifiers (discrete and combined!)
- Most amplifiers have isolated terminals.
- Design an active inductor with 1-2 amplifiers:
  - Various literature is available
  - Active inductor input impedance is heavily influenced by G_m of the amplifiers, the capacitance of its output node, which is also a pole.
  - Pole compensation indirectly means impedance alteration!
- First calculate the transfer function by hand.
- Then form the circuit and connect.
- After that, you can verify the measurements with experimental results.

System-Level Design: The Compensation Challenge (Application based Course project)

- **The Objective**
  - Real-world applications break standard amplifiers. You will be assigned strict system-level specifications. Your task is to diagnose exactly why a standard topology fails under these conditions and implement an advanced compensation strategy to save it.
- **The Engineering Workflow**
  - **Step 1: Diagnose the Failure:** Test a baseline amplifier against your assigned specs to find the exact instability mechanism (e.g., ringing, oscillation, or slow settling).
  - **Step 2: Strategize the Fix:** Select an advanced compensation architecture tailored directly to the failure mechanism.
  - **Step 3: Validate the System:** Prove your newly compensated circuit survives across extreme load conditions and process corners.
- **Target Application Scenarios**
  - **Power & Display Drivers:** Stabilizing the amplifier against massive, unpredictable capacitive loads.
  - **Precision ADC Cores:** Eliminating "slow creeping" settling errors to meet strict high-speed clock deadlines.
  - **RF/Wideband Systems:** Bypassing internal bottlenecks to achieve massive bandwidth without destroying phase margin.

Pins and Switches

- In the circuit diagram, exposed pins are marked by arrowheads.
- There are 21 pins (19 I/O + 2 power i.e. Vdd and GND)
- There are 9 switches inside the circuit

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Exposed pins list

- Gm1 (external signal) Input (2 pins) -> P1 and P2
- Gmff input (2 pins) -> P3 and P5
- Gmff external resistor for value adjustment -> P4
- Off-chip Miller R/C (2 pins) -> P6 and P7
- Off-chip shunt capacitor -> P8
- Gm2 (2nd amplifier) input- pin -> P9
- Off-chip Nested Miller R/C (2 pins) -> P10 and P11 (P10 also acts as output for Gm2 and input for Gm3 while testing)
- Gm3 (3rd amplifier) input- pin -> P12
- Output -> P13
- Vdd pin -> P14
- GND Pin (corner assuming) -> P15
- Inputs to 7-bit decoder (7 pins) -> P16 to P22

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Switches list

- Voltage buffer in series with Miller cap (S1)
- Miller cap shorting switch (S2)
- Uncompensated 2-stage amplifier shorting switch (S3)
- Connect output of Gm2 with input of Gm3 and output of Gm1 with input of Gm2 / connect I/O of both to output pins (S4, S4’)
- Nested Miller/Reverse Nested Miller configuration switcher (S5, S5')
- Feedforward compensation (S6)
- Polarity switcher of Gm2 (S7)
- Polarity switcher of Gm3 (S8)
- Polarity switcher of Gff (S9)

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Control Table

- Pins P13, P14, P15, P16, P17, P18 and P19 are inputs to the 7-bit decoder.
- They drive different configurations of the switches they are connected to.
- The FSM logic is given in the next slide, mapping the bit sequence of P19-P18-P17-P16-P15-P14-P13 to the states of switches S1, S2, S3, S4, S5, S6, S7, S8, S9.

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FSM logic

- When S4 is open, we can toggle S1, S2, S3, S6, S8 or S9, but S5 and S7 will always remain OFF. (2-stage configuration)
  - S1, S2, S3 are mutually exclusive (at max one can be on at one time) but S6 and S8 can be on or off independent of them. (we can use only one compensation network among miller, but pair it with feedforward, toggle it and Gm3 polarity.)
  - only when S6 is on, S9 can be toggled. when S6 if off, S9 can stay only OFF. (only when the Gff path is closed, we can switch its polarity)
- When S4 is closed, we can toggle S1, S2, S5, S6, S7, S8 and S9. S3 will be always off if S4 is closed. (3-stage configuration)
  - we can switch on S1, S2 mutually exclusive (at max one can be on at one time) (only one compensation network among the miller ones can be switched on)
  - independent of S1 or S2, we can toggle S5, S6, S7, S8 freely. (nested miller and feedforward are independent of each other)
  - only when S6 is on, we can toggle S9. when S6 is off, S9 can stay only OFF. (same as above)

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Expressing switch states in terms of pin inputs

- S1 = P13 · (P14' + P15'·P16)
- S2 = P14 · (P13' + P15·P16')
- S3 = P13 · P14 · P15 · P16
- S4 = (P13 · P14)' = P13' + P14'
- S5 = S4 · P15 = (P13' + P14')·P15
- S6 = P13·P14·P17 + S4·P16 = P13·P14·P17 + (P13'+P14')·P16
- S7 = S4 · P17 = (P13' + P14')·P17
- S8 = P18
- S9 = S6 · P19 = [P13·P14·P17 + (P13'+P14')·P16]·P19

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Expressing pins in terms of switches

- P13 = S4' + S1
- P14 = S4' + S2
- P15 = S4'.(S2 + S3) + S4.S5
- P16 = S4'.(S1 + S3) + S4.S6
- P17 = S4'.S6 + S4.S7
- P18 = S8
- P19 = S9 (valid under FSM rule: S9 ≤ S6)

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FSM logic: Truth table—S4 open (P13=P14=1) [2-stage]

Compensation selected (P15,P16)	Feedforward (P17)	Gm3 polarity (P18)	S1 (Miller with voltage buffer)	S2 (Normal Miller)	S3 (2 stage uncompensated)	S5 (Nested)	S6 (FF path)	S7 (Nested2)	S8 (Gm3)	S9 (FF polarity)
None (0,0)	off (0)	normal (0) / inv (1)	off	off	off	off	off	off	P18	off
None (0,0)	on (1)	normal / inv	off	off	off	off	on	off	P18	0
Miller with voltage buffer (0,1)	off (0)	normal / inv	on	off	off	off	off	off	P18	off
Miller with voltage buffer (0,1)	on (1)	normal / inv	on	off	off	off	on	off	P18	0
Normal Miller (1,0)	off (0)	normal / inv	off	on	off	off	off	off	P18	off
Normal Miller (1,0)	on (1)	normal / inv	off	on	off	off	off	off	P18	0
2 stage uncompensated (1,1)	off (0)	normal / inv	off	off	on	off	off	off	P18	off
2 stage uncompensated (1,1)	on (1)	normal / inv	off	off	on	off	off	off	P18	0

- **S_5 = 0, S_7 = 0 always** (forced off in 2-stage).
- **S_9 is active (= P19) only when feedforward is on** (P17=1); otherwise it is forced to 0.
- **(P15, P16) select the Miller compensation network; they are mutually exclusive** (at most one can be 1 at any time; the state P15=0, P16=0 represents "no compensation selected").
- **P18 sets the Gm3 polarity** (0 = normal, 1 = inverted).

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FSM logic: Truth table—S4 closed [3-stage]

S1	S2	S3	S5	S6	S7	S8	S9
P13	P14	0	P15	P16	P17	P18	P16 • P19

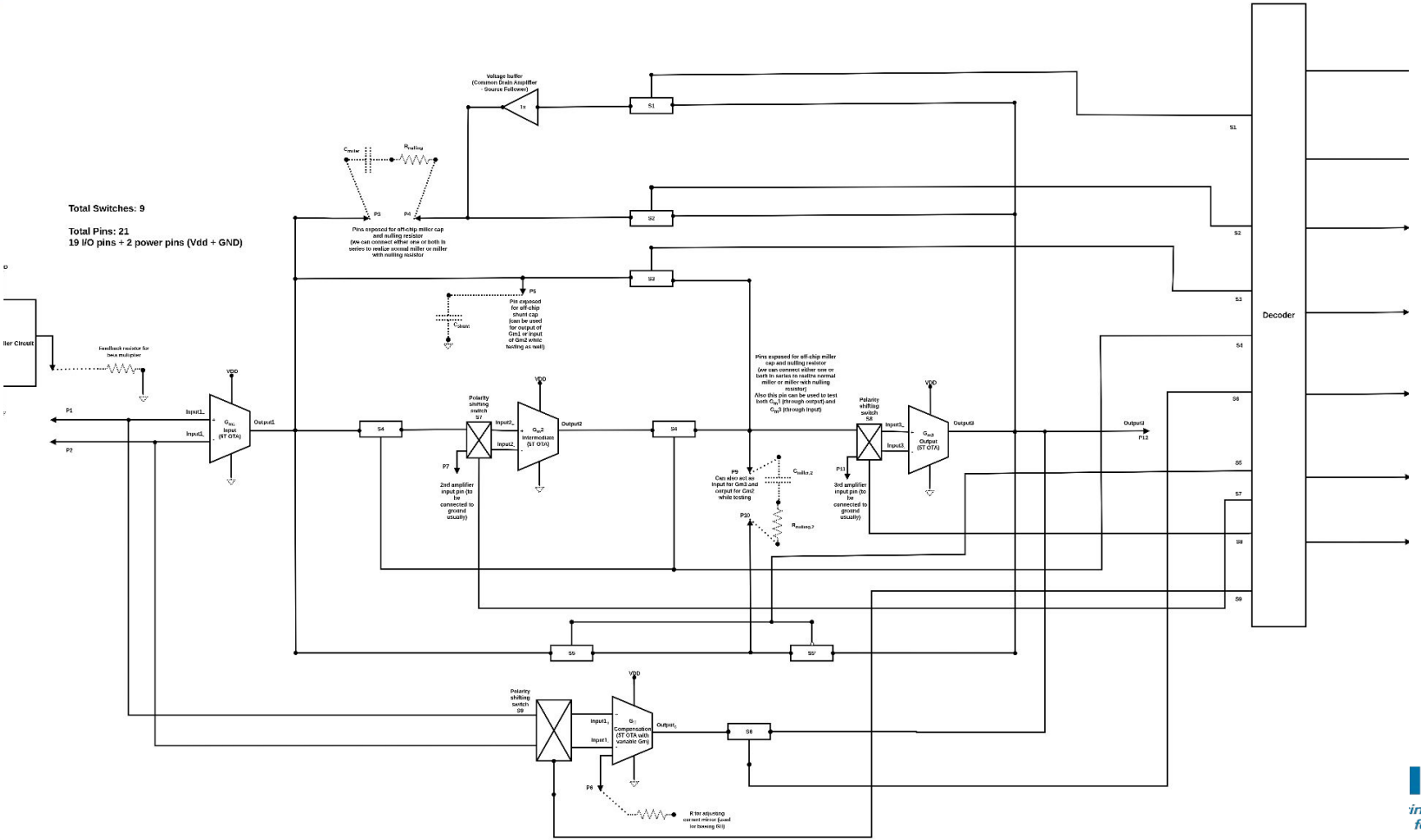
- **S_1 = 0, S_2 mutually exclusive** (at most one can be 1 at any time; P13 and P14 cannot both be 1 simultaneously; the state P13=0, P14=0 represents "no Miller compensation selected").
- **S_3 = 0 always** (no third Miller network in 3-stage).
- **S_5, S_6, S_7, S_8 are independently controllable** (nested Miller capacitor, feedforward path, second nested Miller, and Gm3 polarity are directly controlled by P15, P16, P17, P18 respectively).
- **S_9 is active (= P19) only when feedforward is on** (P16=1); otherwise it is forced to 0.

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Circuit Diagram
(Of the full chip)

https://drive.google.com/file/d/1K-ecmafkDB78uYrOsWLTLrin3UoFuY0m/view?usp=sharing

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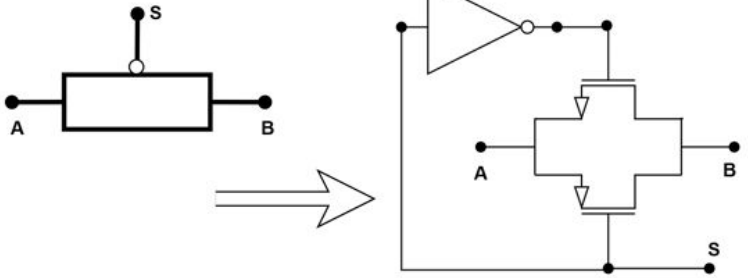


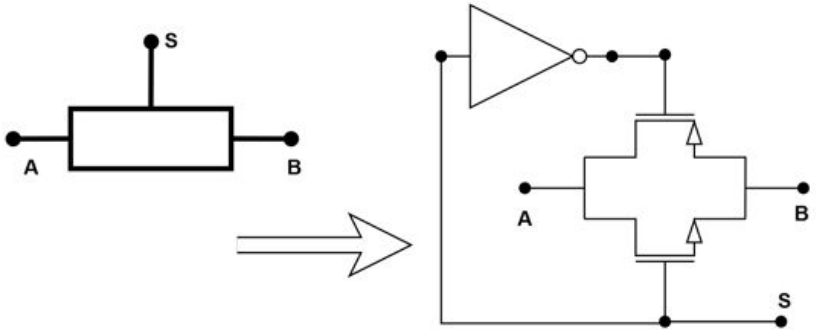
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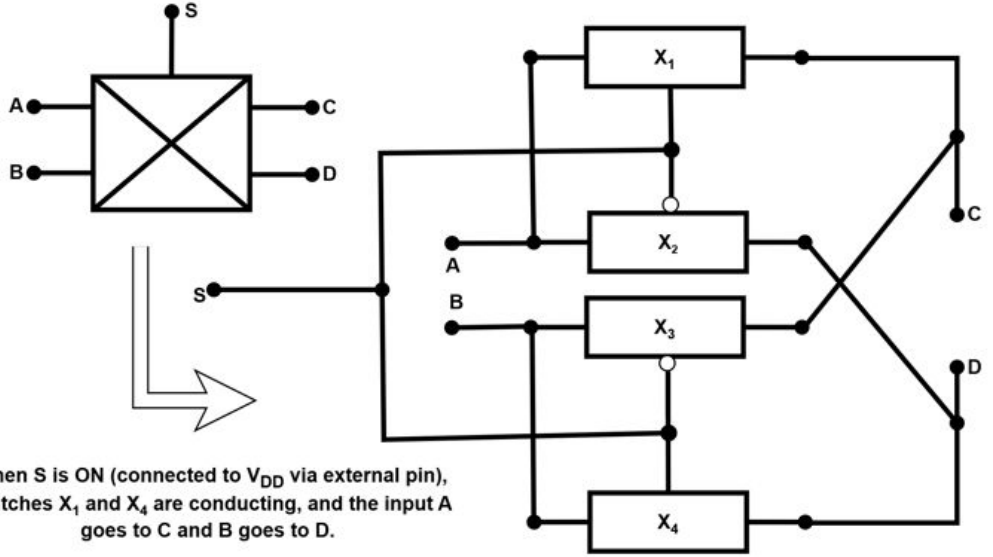
Subcomponents

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Switches







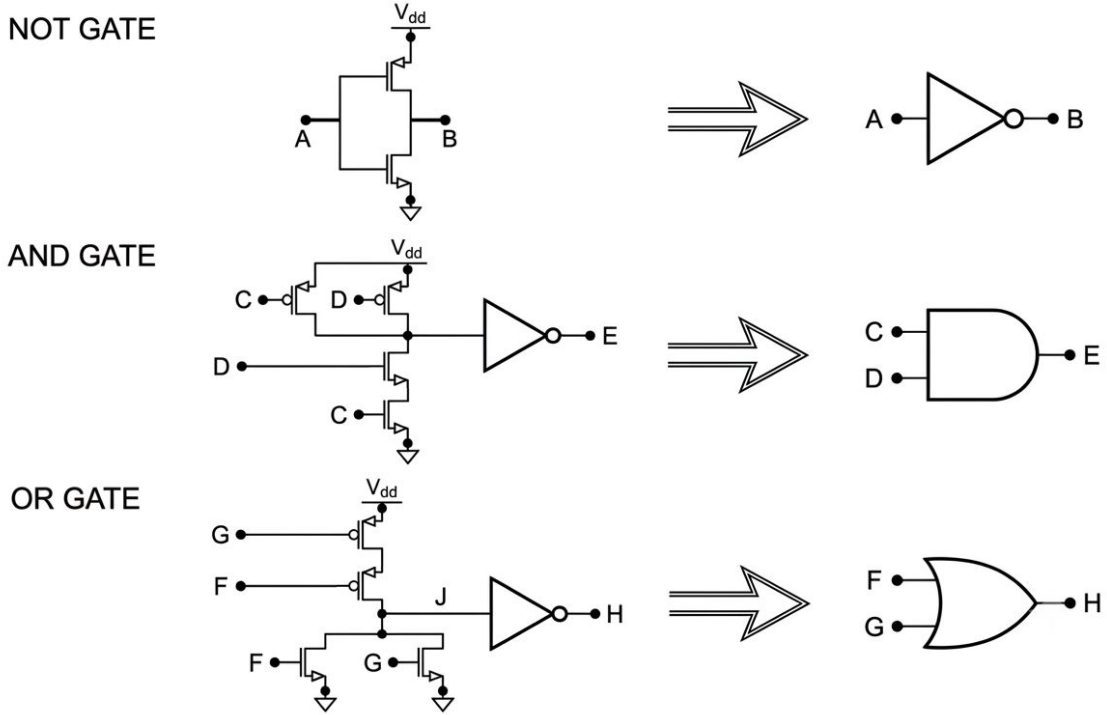
When S is ON (connected to V_{DD} via external pin), switches X_1 and X_4 are conducting, and the input A goes to C and B goes to D.

When S is OFF (connected to ground via external pin), switches X_2 and X_3 are conducting, and the input A goes to D and B goes to C.

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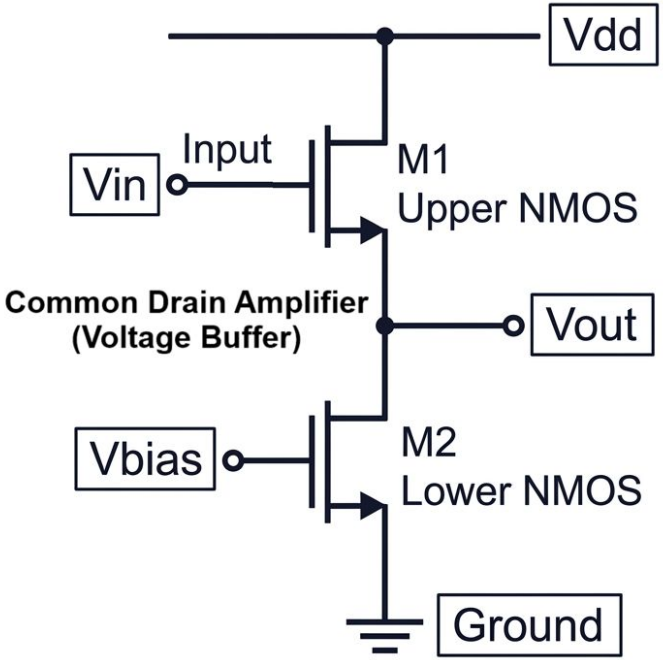
Gates

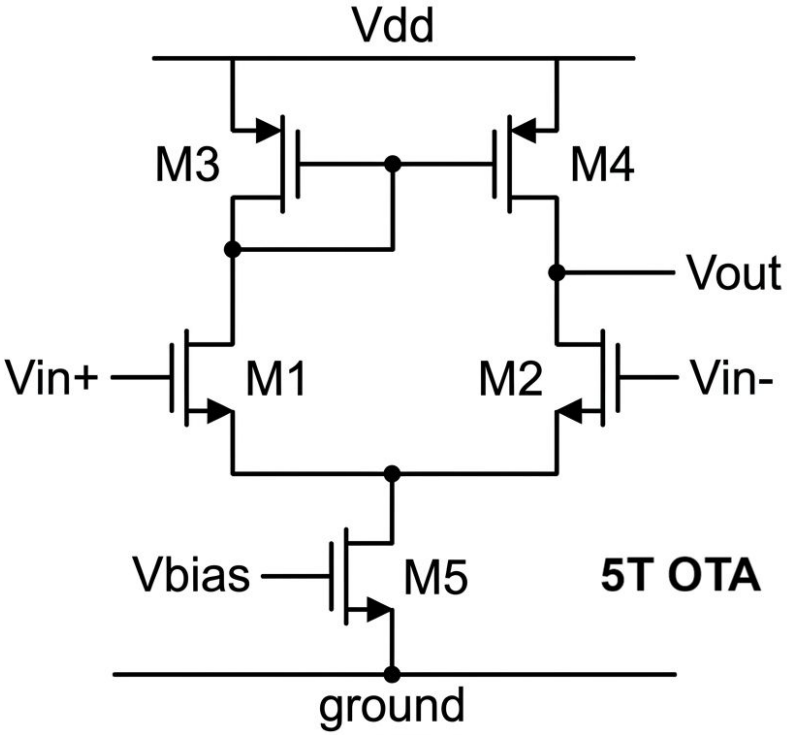
CMOS Logic Gates: Transistor Circuitry and Symbolic Representation



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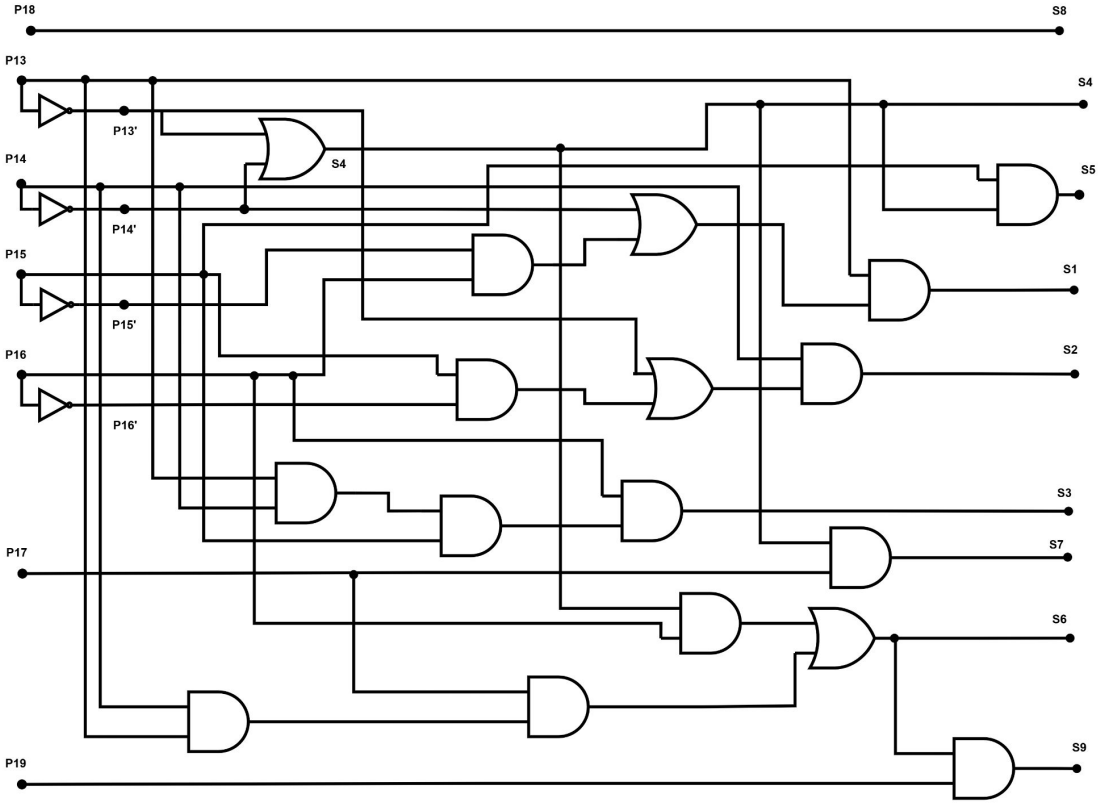
Amplifiers





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Decoder

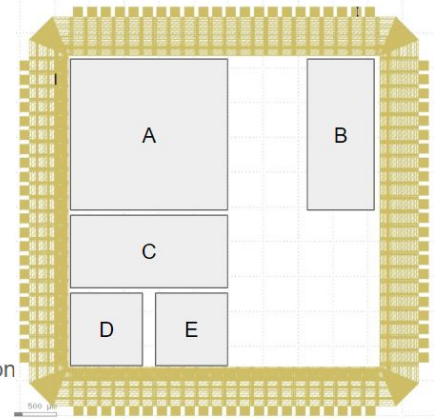


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Area Estimate

- We are using Padframe A!
- As a rule of thumb, metal (dummy + routing) occupies roughly 8-9 times the silicon area in 180nm technology node.
- Total area is 1,250,000 µm²
- Therefore, silicon (amplifiers + switches) should occupy 10% of the floor area (125,000 µm²)
- We should be sizing them accordingly!

- 88 pins
- Total project area 2235um x 2235um
- 5 block configurations
  - A: 1/4 - 22 pins (4 possible locations)
  - B: 1/8 - 16 pins (4 possible locations)
  - C: 1/8 - 6 pins (8 possible locations)
  - D: 1/16 - 10 pins (4 possible locations)
  - E: 1/16 - 6 pins (8 possible locations)
- Separate power for each block
- Chip wide common ground
- Each project specifies i/o pad types
- Placement fixed for each block
- Designs connect to pads when placed
- Smaller projects may be combined into one block. Needs a person to manage block integration



Simulation

- We have simulated the following using rigorous reasoning, aided by the gm/Id method:
  - 5T OTA
  - CMOS switch
- The following simulations are in progress as we are writing the scripts for their gm/Id analysis:
  - 6T OTA
  - Voltage buffer
- We aim to complete all the simulations and integration by 30th, learn layout tools, and will begin layout on 1st!
- The reasoning has been done, so simulations would take very less time!

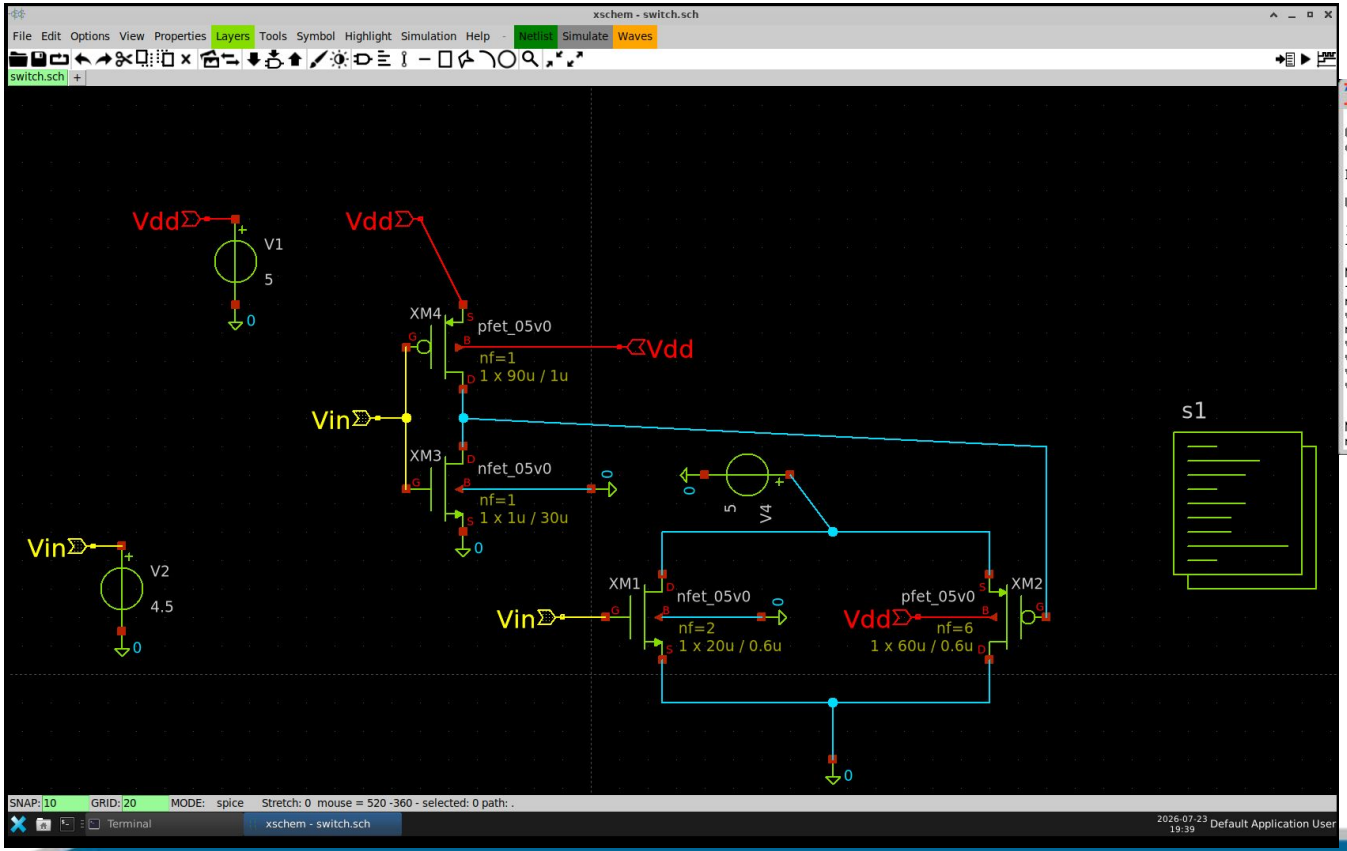
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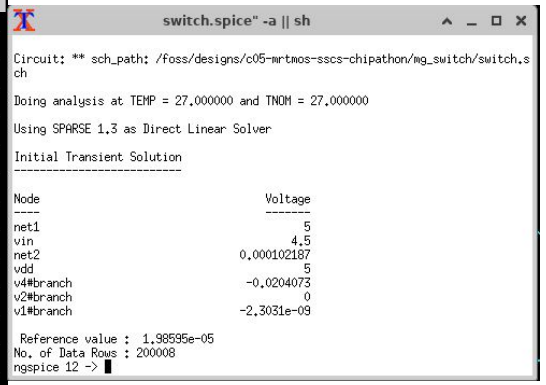
Switch Simulation

- The ON current is coming around 20mA and the OFF current is of the order of nA
- This means that the switch conducts fairly (has around 250 ohm ON resistance) and blocks very well.
- Operating points are shown in the next slide.

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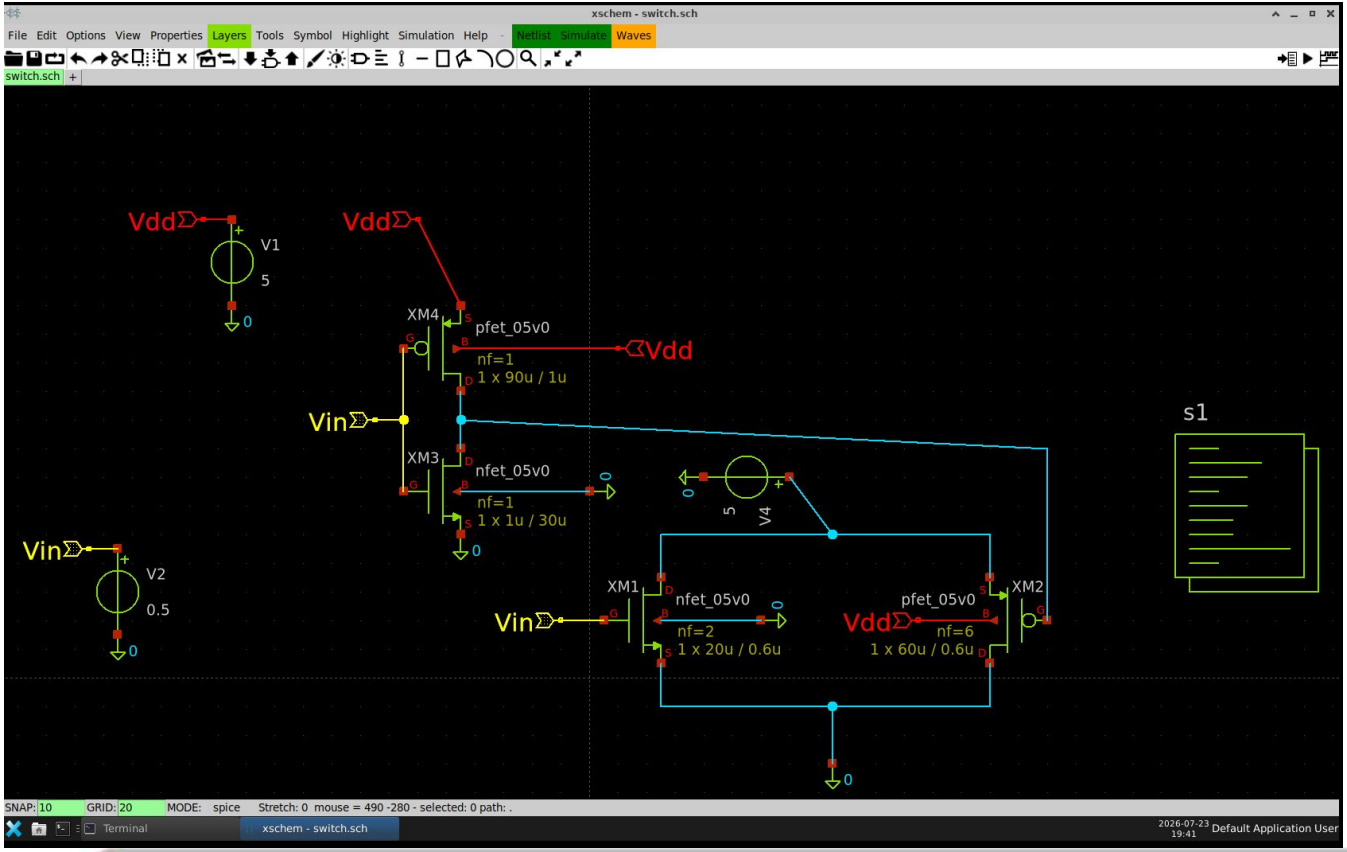
Switch Simulation (ON state, conducting current)

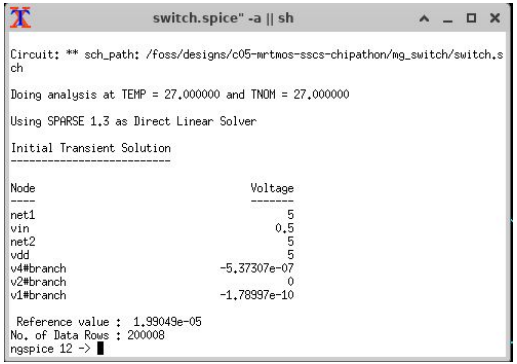




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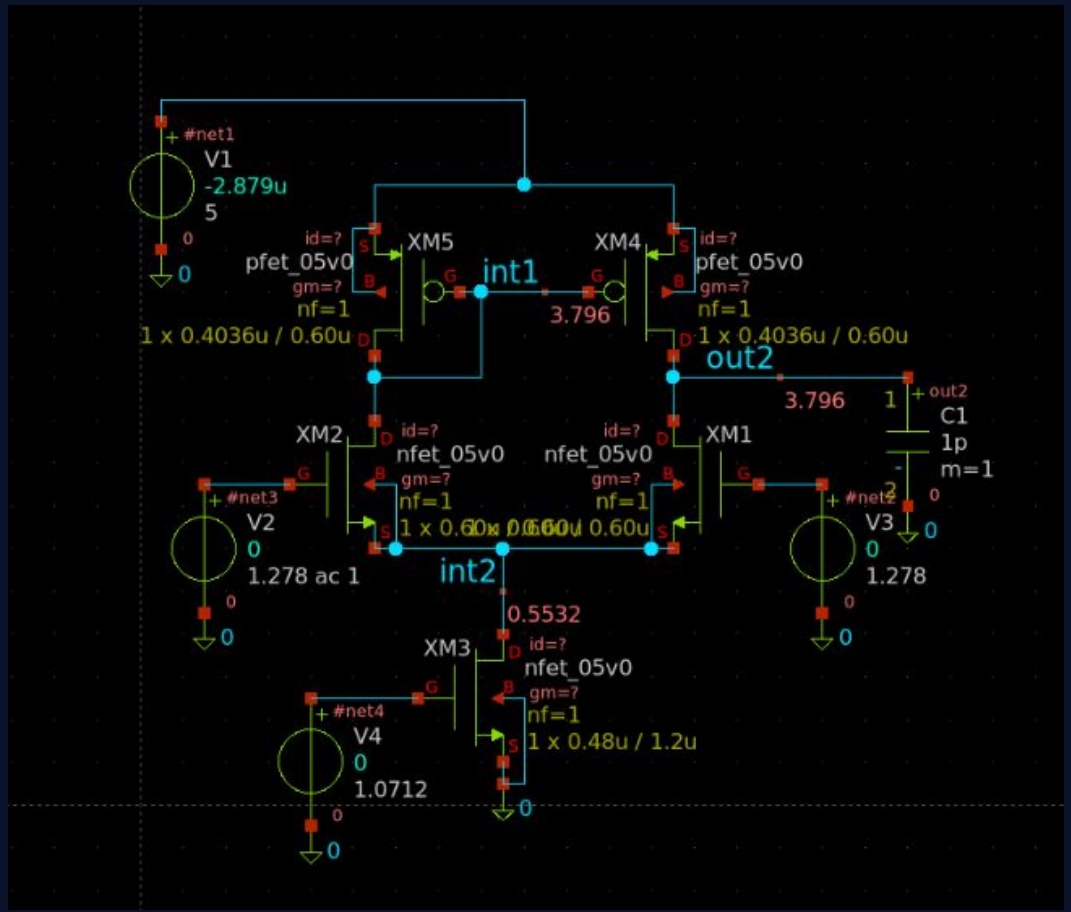
Switch Simulation (OFF state, blocking current)





5T OTA Simulation

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Circuit Schematic

Operational Transconductance Amplifier (OTA) design with active loads and ideal bias circuitry.

Simulations

Five-transistor OTA core configured for AC analysis, evaluating node voltages at **int1**, **int2**, and **out2**.

Model Libraries

```
.lib
/foss/pdks/ciel/gf180mcu/versions/7b70722e33c03
fcb5dabcf4d479fb0822d9251c9/gf180mcuD/libs.tech
/ngspice/sm141064.ngspice typical
```

```
.include
/foss/pdks/ciel/gf180mcu/versions/7b70722e33c03
fcb5dabcf4d479fb0822d9251c9/gf180mcuD/libs.tech
/ngspice/design.ngspice
```

✓ DC operating points annotated

```
.control
        save all
        op
        remzerovec
        write 1_OTA_1.raw
.endc
.options savecurrents
```

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Methodology: Murmann LUT-Based gm/Id Sizing

Choosing gm/Id Targets

1. Differential Pair: 12 V^{-1}

Provides the high transconductance needed for target bandwidth without pushing the devices so wide that parasitic capacitance degrades speed.

As gm/Id goes up, ID/W goes down; means W shoots up !

2. PMOS Active Load & Tail: 5 V^{-1}

Pushes devices deep into saturation (high V_DSAT) to deliver high output resistance for optimal gain and lower noise as transconductance is scaled.

MINIMUM WIDTH (W) - 5V DEVICES

0.3 µm (PMOS & NMOS)

MINIMUM CHANNEL LENGTH (L)

PMOS: 0.5 µm

NMOS: 0.6 µm

THRESHOLD VOLTAGES (Vth)

Vth, n = 0.70 V

Vth, p = 0.83 V

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Initial Sizing & Design Iterations

1. Design Targets

FREQUENCY & LOAD

fu = 2.5 MHz

Buffer over 1MHz target

CL = 1 pF

CALCULATED PARAMETERS

gm = 16 µS

gm = 2*pi*fu*CL

Id = 1.34 µA

Itail = 2.68 µA

2. Channel Length & Gain

INITIAL RANDOM SIZING

L values chosen randomly:

- Kept most L's at minimum
- Tail source L = 1.2 µm (for high ro)
- **Achieved DC gain of ~32 dB**

THE RECOMMENDED WAY

Choose **L** systematically based on the **gm/gds vs. gm/Id** plots to optimize gain and output resistance.

3. Vds & Common Mode

UNSATISFIED ASSUMPTION

Assumed Vds of **2V** (diff pair) and **0.5V** (tail) to keep output common mode at 2.5V.

Resulted in output CM of 3.796V

CORRECT IMPLEMENTATION

Connect the gate of XM3 to a diode-connected NMOS reference transistor.

Drive with an ideal **2.68 µA** current source (`isource.sym`).

Creates a proper current mirror that holds tail current perfectly at **2.68 µA** regardless of VDS swings.

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Sizing Lookup & Parameter Extraction

1. Vgs Lookup

Extract Vgs of the differential pair and tail source based on chosen gm/Id and L.

```
# Load technology parameters
nmos_5v = lk('nfet_05v0.mat')

# Define sizing targets
target_gmid = 12
length_um = 0.6
vds_operating = 2

# Extract required Vgs
vgs_req = db.lookupVGS(
  gm_id=target_gmid,
  L=length_um,
  VDS=vds_operating
)
```

OTHER VALUABLE LOOKUPS

gm/Id v/s Id/W, gm/gds, fT, Vgs, Cgg/W, Cgd/W, Cdd/W

2. Width (W) Sizing

Next, resolve the width by looking up Id/W v/s gm/Id under operating conditions.

```
# Lookup current density Id/W
id_w_val = db.lookup(
  'ID W',
  gm_id=target_gmid,
  L=length_um,
  VDS=vds_operating
)
```

Found W from here!

Width parameters calculated and locked in dynamically.

Design Phase Completed

The circuit schematic has been successfully generated!

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Mismatch & Parameter Deviations

Due to **Vds mismatch** between design assumptions and actual simulation, the tail current (`Itail`) shifted to **2.88 µA**.

1. Diff Pair Gm

ACTUAL VALUE

17.9 µS

EXPECTED TARGET

16.0 µS

2. Tail Source Gm

ACTUAL VALUE

13.26 µS

EXPECTED TARGET

14.4 µS

3. Active Load PMOS Gm

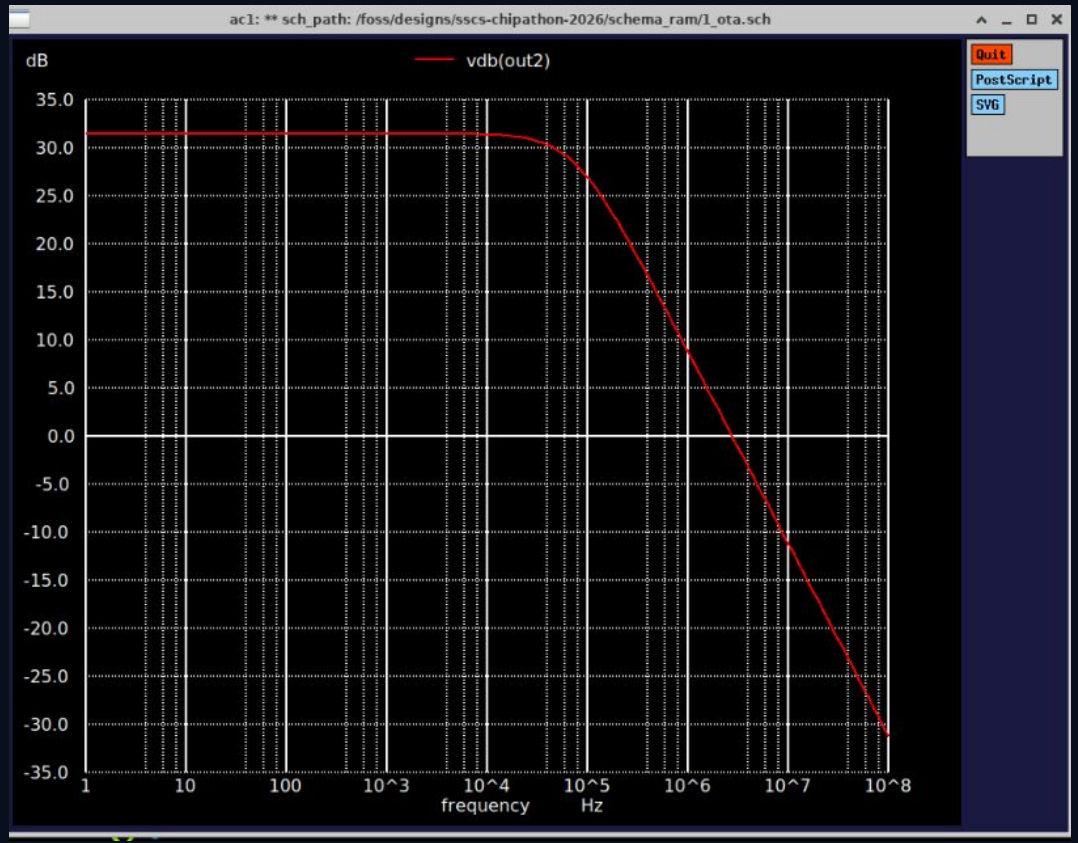
ACTUAL VALUE

6.29 µS

EXPECTED TARGET

7.2 µS

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Simulation Output

Frequency response plot showing gain and phase behavior of the OTA circuit.

NGSPICE TERMINAL COMMAND

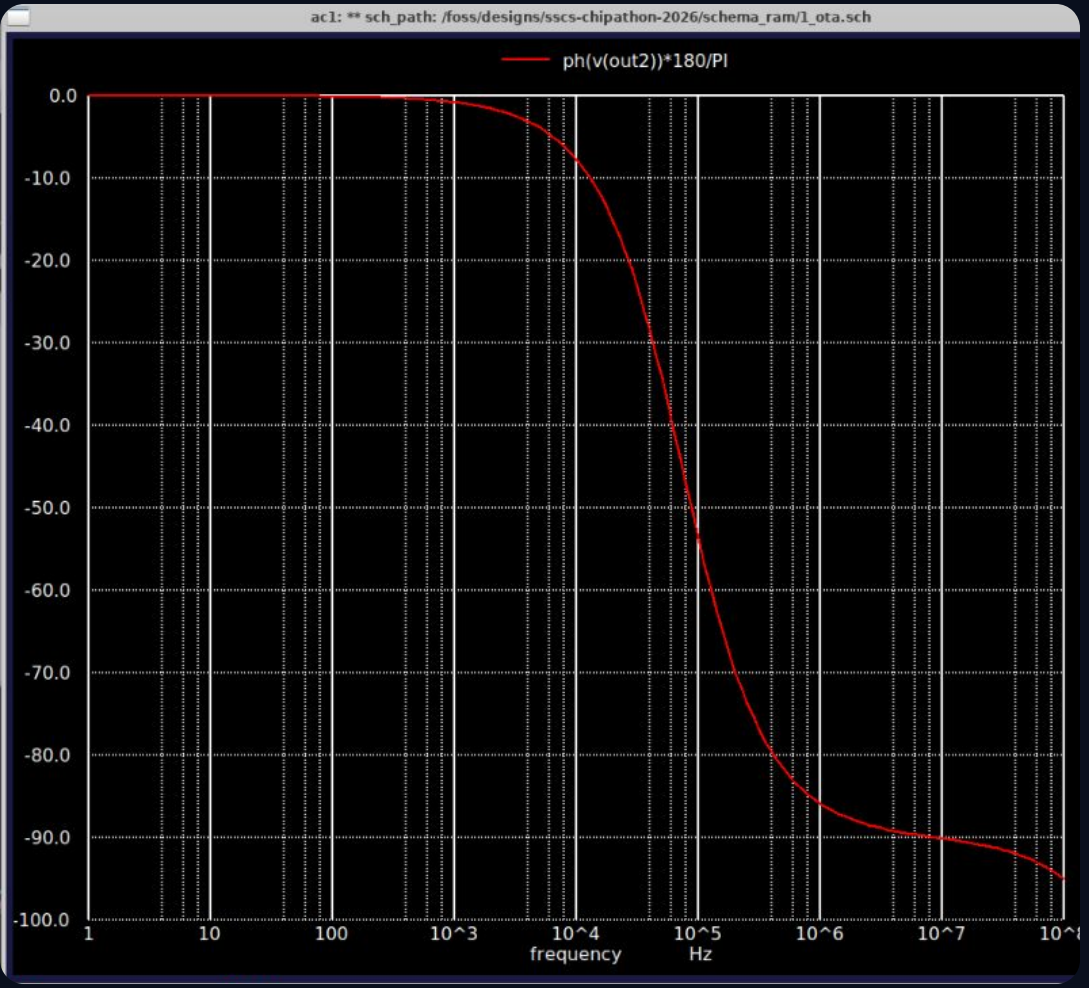
```
plot vdb(out2) xlog
```

```
Low Freq Gain =32.16dB
fu= 2.73 MHz
CL used = 1pF
```

Put this in the simulations code block to run the ac simulation

```
.control
  save all
  ac dec 20 1 100Meg
  remzerovec
  write 1_OTA_ac.raw
.endc
```

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Phase Response

Frequency response plot analyzing the phase behavior and stability margin of the OTA circuit.

Command Used

```
plot ph(v(out2))*180/PI xlog
```

Analysis Metrics

Phase at 2.73 MHz

At unity gain frequency (gain = 0 dB)

-87.94°

Phase Margin

So PM = 92.06°

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References

1. https://www.crowdsupply.com/wafer-space/gf180mcu-run-1/updates/mosbius
2. Behzad Razavi, Design of Analog CMOS Integrated Circuits
3. R. Jacob Baker, CMOS: Circuit Design, Layout, and Simulation
4. Gray and Meyer, Analysis and Design of Analog Integrated Circuits
5. https://milindsweb.amved.com/docs/Ahuja_Compensation_Paper.pdf
6. https://pallen.ece.gatech.edu/Academic/ECE_6412/Spring_2004/L130-OpAmpCompII(2UP).pdf
7. https://en.wikipedia.org/wiki/Breadboard
8. https://www.analog.com/en/resources/analog-dialogue/studentzone/studentzone-november-2016.html
9. https://www.microcircuitsjournal.com/article/80/6-1-4-976.pdf
10. https://electronics.stackexchange.com/questions/302850/what-order-of-magnitude-should-i-expect-the-parasitic-capacitance-to-be-for-a-so
11. https://www.deepdyve.com/lp/unpaywall/parasitic-capacitance-effects-in-breadboard-circuits-operating-above-1-px0TTlA2kh
12. **IIT Delhi courses class notes:**
    a. Analog Integrated Circuits (ELL735)
    b. Advanced Analog Circuits: Analysis and Design of Op-Amps (ELL830)
       Mixed Signal Circuit Design (ELL731)
    c. Special Topics in RF Design - PLL Design (ELL834)
    d. Semiconductor Memory Design (ELL749)
13. Claude and Gemini for extraction and fact-checking and generating some clean diagrams

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Some Remarks

- Simulations ~~will be done~~ is being done after we size the transistors based on the gm/Id values (in progress)
- Didn’t want to just blatantly simulate or calculate values using AI, as we won’t learn the technique then, and we will be taping out on silicon, so we should know everything in and out.
- Also, AI is not that trustworthy for calculations, designing and numbers in electronics, based on our experience, so we prefer to do everything by hand. AI is mainly used for fact-checking and rewriting or remaking diagrams to make them cleaner.
- Didn’t want to use AI for cognitive outsourcing, but rather as a verification/fact-checking tool.
- Suggestions and questions are welcome :)
- However, we would want the answer to some questions:
  - Are we heading in the right direction?
  - Is the speed fine, or do we need to speed up?
  - What all are the shortcomings/scopes of improvement?

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Thank you :)

Questions?

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